

Supplemental Data: Rule-Unit Anatomy, the Retrieval Loop and Provision Provenance

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This supplement accompanies “Regulation-Grounded Visual Compliance Screening of Construction Sites: Rule Applicability Retrieval, Not Visual Judgement, Governs Accuracy”. It reports the field anatomy of a single provision unit (Fig. S1), the pseudocode of the tool-calling retriever (Algorithm S1) and the per-provision provenance of the 42-provision library (Table S1). The same information is machine-readable in the public artifact repository (Zenodo 10.5281/zenodo.21535054). Nothing here is needed to follow the argument of the manuscript; this file records the detail a reader would need in order to audit or reproduce it.

RULE-UNIT ANATOMY

Fig. S1 shows the typed fields of one provision unit and the auditable chain they expand into. The example is R-BHV-002, the helmet provision.

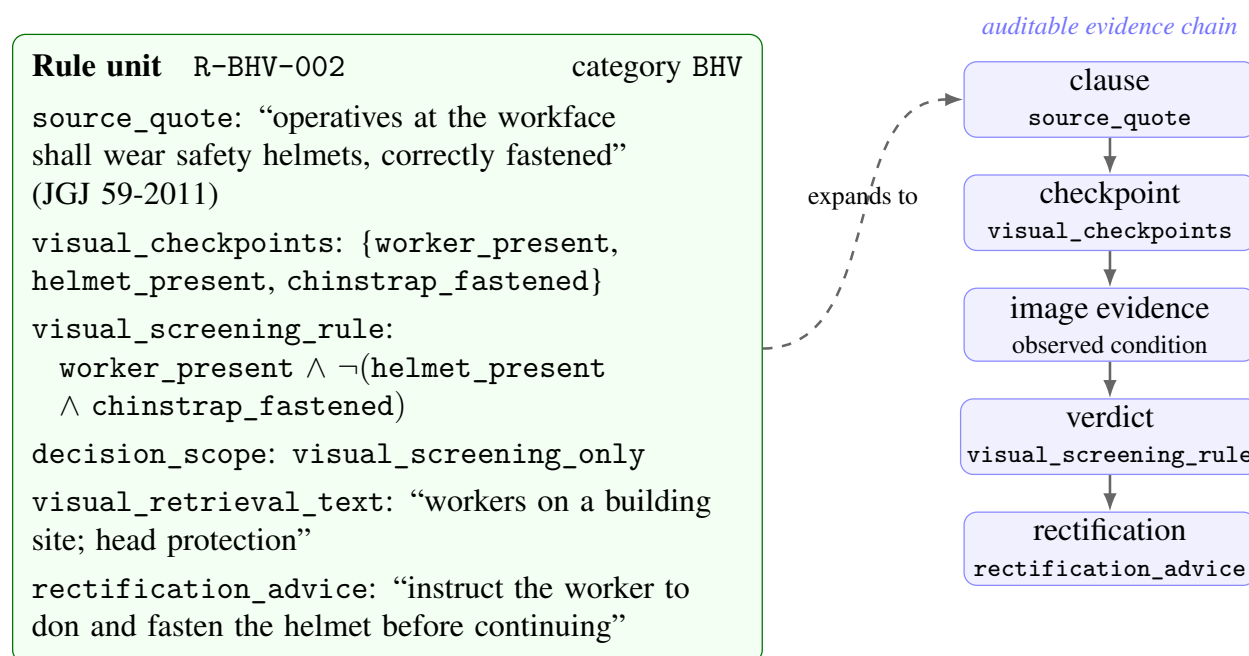


FIG. S1. Anatomy of a provision unit (example R-BHV-002, the helmet provision). The typed fields (left) carry provenance, atomic perception targets, a Boolean decision formula with an explicit decision scope, retrieval text and remediation. Because the fields are separable, one hit expands deterministically into the five-link auditable chain (right) that a reviewer inspects.

RETRIEVAL LOOP

Algorithm S1 specifies the Stage-2 tool-calling retriever of the manuscript’s “Rule retrieval: fixed and tool-calling” section. R3 and R4 differ only in the observation O : R4 additionally receives the raw image on every turn.

Algorithm S1: Bounded tool-calling rule retrieval (R3/R4) with anytime termination. $\text{FORCESUBMIT}(H)$ takes one extra turn with the tool pinned to submit and, if that still yields no known rule_id, falls back to the ids in H ranked most-hit-first. An iteration-cap hit, a stall, a truncated turn or an unusable submission thus degrades to a best-effort candidate list rather than an empty one.

Input : observation O (cached facts; plus the raw image for R4); provision corpus $\mathcal{C}$; iteration budget N ; stall limit L .

Output: a ranked list of candidate rule_ids.

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1  $H \leftarrow \{\}$  // surfaced rule_id  $\mapsto$  cumulative grep hits
2  $P \leftarrow \emptyset$ ;  $s \leftarrow 0$ ;  $e \leftarrow 0$  // patterns run; stall counter; empty turns
3 for  $i \leftarrow 1$  to  $N$  do
4    $force \leftarrow (i = N)$  // final turn pins the tool to submit
5    $r \leftarrow \text{Model}(\text{history}, \text{tools}, \text{force})$ 
6   if  $r$  emits no tool call then
7      $e \leftarrow e + 1$ 
8     if  $r$  truncated or  $e \geq 2$  or  $force$  then
9       return  $\text{ForceSubmit}(H)$ 
10    append retry nudge // try once more
11  else
12    foreach tool call  $c \in r$  do
13      if  $c = \text{submit}(\text{ids})$  then
14         $K \leftarrow \{x \in \text{ids} : x \in \mathcal{C}\}$  // keep known ids
15        if  $K \neq \emptyset$  then
16          return  $\text{Rank}(K)$ 
17        else
18          return  $\text{ForceSubmit}(H)$  // unusable submit
19      else if  $c = \text{grep}(p)$  then
20         $\text{hits} \leftarrow \mathcal{C}.\text{grep}(p)$  // matching rule_ids + snippet
21         $\text{new} \leftarrow$  number of ids in  $\text{hits}$  not yet in  $H$ ; update  $H$ 
22        if  $p \in P$  or  $\text{new} = 0$  then
23           $s \leftarrow s + 1$ 
24        else
25           $s \leftarrow 0$ 
26         $P \leftarrow P \cup \{p\}$ 
27      else
28        return the full provision detail of  $x$  //  $c = \text{read}(x)$ 
29    if  $s \geq L$  then
30      return  $\text{ForceSubmit}(H)$  // marginal-gain early stop
31 return  $\text{ForceSubmit}(H)$  // safety net; unreachable when  $N \geq 1$ 

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PER-PROVISION PROVENANCE

Table S1 gives the provenance of all 42 provisions; its caption defines the six columns. The standard designations are **GB 55034** (MOHURD 2022), **GB 12523** (MEE 2025), **JGJ 59** (MOHURD 2011), **JGJ 80** (MOHURD 2016), **Hazard 2024** (MOHURD 2024), **SpecOps 2008** (MO-

HURD 2008) and **SZ 2018** (SZHCB 2018). Some civilized-construction thresholds are Shenzhen local reference values rather than national mandatory limits, and all source_quote text in the English runtime library is an author translation, not a verbatim quotation.

TABLE S1. Per-provision provenance of the 42-provision library. *Cat.*: OPN openings, EDG edges, BHV behaviour and protective equipment, CIV civilized-site. *Standard(s)* names the governing document, cited above; “—” marks a provision with no single governing clause. *Prov.*: std., the provision transcribes a clause of that standard; eng., an engineering operationalization, whether or not a standard is also cited; acc. and mgmt., additionally derived from accident-investigation findings or from site management requirements. *Pred.*: clause when visual_screening_rule is identical to the transcribed normative_rule and needs no external field, proxy otherwise. *Drop*: normative fields the clause requires that a single image cannot supply. *Scope*: the library’s decision_scope. These last three are read directly off the frozen library, so they cannot drift from the provisions the pipeline runs.

Rule	Cat.	Standard(s)	Prov.	Pred.	Drop	Scope
BHV-001	BHV	GB 55034; JGJ 80	eng.	proxy	2	screen
BHV-002	BHV	GB 55034	eng.; acc.	clause	0	normative
BHV-004	BHV	JGJ 80	std.	clause	0	normative
BHV-005	BHV	JGJ 80	std.	clause	0	normative
BHV-006	BHV	SpecOps 2008	std.	proxy	2	external
BHV-007	BHV	—	acc.	proxy	1	screen
BHV-008	BHV	JGJ 80	acc.	proxy	1	screen
BHV-009	BHV	Hazard 2024	acc.	proxy	2	screen
BHV-010	BHV	SZ 2018	std.	proxy	1	screen
BHV-011	BHV	Hazard 2024	acc.	proxy	1	external
BHV-012	BHV	—	acc.; mgmt.	clause	0	normative
BHV-013	BHV	JGJ 80	eng.	clause	0	normative
CIV-001	CIV	SZ 2018; JGJ 59	std.	proxy	4	screen
CIV-002	CIV	SZ 2018; JGJ 59	std.	clause	0	normative
CIV-003	CIV	SZ 2018; JGJ 59	std.	proxy	2	screen
CIV-004	CIV	SZ 2018	std.	clause	0	screen
CIV-005	CIV	SZ 2018	std.	proxy	5	screen
CIV-006	CIV	SZ 2018	std.	proxy	6	external
CIV-007	CIV	SZ 2018	std.	proxy	4	screen
CIV-008	CIV	SZ 2018	std.	proxy	6	external
CIV-009	CIV	GB 12523	std.	proxy	7	external
CIV-010	CIV	JGJ 59	std.	proxy	8	screen
CIV-011	CIV	SZ 2018; JGJ 59	std.	proxy	1	screen
CIV-012	CIV	SZ 2018; JGJ 59	std.	proxy	1	screen
CIV-013	CIV	SZ 2018	std.	proxy	3	screen
CIV-014	CIV	SZ 2018; JGJ 59	std.	clause	0	normative
CIV-015	CIV	SZ 2018; JGJ 59	std.	proxy	1	screen
CIV-016	CIV	—	eng.	proxy	0	normative
CIV-018	CIV	SZ 2018; JGJ 59	std.	clause	0	normative
EDG-002	EDG	JGJ 80	std.	clause	0	normative
EDG-003	EDG	JGJ 80	std.	clause	0	normative
EDG-004	EDG	JGJ 80	std.	proxy	2	screen
EDG-005	EDG	Hazard 2024; SZ 2018	std.	clause	0	normative
EDG-006	EDG	JGJ 80	std.	proxy	1	screen
EDG-007	EDG	JGJ 80	std.	proxy	8	screen
EDG-011	EDG	JGJ 80	eng.	proxy	1	screen
EDG-012	EDG	SZ 2018	eng.	clause	0	normative
OPN-001	OPN	SZ 2018; JGJ 80	std.	proxy	5	screen
OPN-004	OPN	JGJ 80	std.	proxy	4	screen
OPN-005	OPN	SZ 2018; JGJ 80	std.	proxy	8	screen
OPN-009	OPN	SZ 2018; JGJ 80	std.	clause	0	normative
OPN-011	OPN	JGJ 80	eng.	clause	0	normative

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